Homework Assignment 5

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Chapter 10, Question 4:

For each natural number n, let $S(n)$ denote the statement that

$$1 \cdot 2 + 2 \cdot 3 + ... + n(n+1) = \frac{n(n+1)(n+2)}{3}.$$

(Basis Step) Then, $S(1)$ states that $1(1+1) = \frac{1 \cdot 2 \cdot 3}{3}$. This simplifies to $2 = 2$, so $S(1)$ is true.

(Inductive Step) For an arbitrary natural number k, assume $S(k)$. That is,

$$1 \cdot 2 + 2 \cdot 3 + ... + k(k+1) = \frac{k(k+1)(k+2)}{3}.$$

Adding $(k+1)(k+2)$ to both sides, we have

$$\begin{aligned} 1 \cdot 2 + 2 \cdot 3 + ... + k(k+1) + (k+1)(k+2) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\ &= \frac{(k+1)(k+2)(k+3)}{3}. \end{aligned}$$

That is, $S(k+1)$ is true. Thus, we have shown that $S(k)$ implies $S(k+1)$.

Therefore, by the Principle of Mathematical Induction, $S(n)$ is true for $n \in N$, establishing the identity

$$1 \cdot 2 + 2 \cdot 3 + ... + n(n+1) = \frac{n(n+1)(n+2)}{3}.$$

Chapter 10, Question 12:

For each integer $n \geq 0$, let $S(n)$ denote the statement that

$$9 \mid (4^{3n} + 8).$$

(Basis Step) Then, $S(0)$ states that $9 \mid 4^{3(0)} + 8$. Since $4^0 + 8 = 9 = 1 \cdot 9$, and 1 is an integer, it follows that $9 \mid 4^{3(0)} + 8$ is true. So $S(0)$ is true.

(Inductive Step) For an arbitrary non-negative integer k, assume $S(k)$. That is,

$$9 \mid (4^{3k} + 8).$$

Then there exists an integer x such that

$$4^{3k} + 8 = 9x, \text{ where } x \text{ is integer.}$$

Multiplying both sides by 64, we have

$$\begin{aligned} 4^{3k+3} + 512 &= 576x, \\ 4^{3(k+1)} + 8 &= 576x - 504, \\ \text{and } 4^{3(k+1)} + 8 &= 9(64x + 56). \end{aligned}$$

Let $w = 64x + 56$. Then,

$$4^{3(k+1)} + 8 = 9w, \text{ where } w \text{ is a integer.}$$

So by definition of divisibility, we have $9 \mid 4^{3(k+1)} + 8$. That is, $S(k+1)$ is true. Thus, we have shown that $S(k)$ implies $S(k+1)$.

Therefore, by the Principle of Mathematical Induction, $9 \mid (4^{3n} + 8)$ for every integer $n \geq 0$.

Chapter 10, Question 22:

For each natural number n, let $S(n)$ denote the statement that

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right) \ldots \left(1 - \frac{1}{2^n}\right) \geq \frac{1}{4} + \frac{1}{2^{n+1}}.$$

(Basis Step) Then, $S(1)$ states that $(1 - \frac{1}{2}) \geq \frac{1}{4} + \frac{1}{2^{1+1}}$. This simplifies to $\frac{1}{2} \geq \frac{1}{2}$, so $S(1)$ is true.

(Inductive Step) For an arbitrary natural number k, assume $S(k)$. That is,

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right) \ldots \left(1 - \frac{1}{2^k}\right) \geq \frac{1}{4} + \frac{1}{2^{k+1}}.$$

Multiplying both sides by $(1 - \frac{1}{2^{k+1}})$, we have

$$\begin{aligned}\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)...\left(1-\frac{1}{2^k}\right)\left(1-\frac{1}{2^{k+1}}\right) &\geq \left(\frac{1}{4}+\frac{1}{2^{k+1}}\right)\left(1-\frac{1}{2^{k+1}}\right)\\ &\geq \frac{1}{4}+\frac{1}{2^{k+1}}-\frac{1}{2^{k+3}}-\frac{1}{2^{2k+2}}\\ &\geq \frac{1}{4}+\frac{4}{2^{k+3}}-\frac{1}{2^{k+3}}-\frac{1}{2^{2k+2}}\\ &\geq \frac{1}{4}+\frac{2}{2^{k+3}}+\frac{1}{2^{k+3}}-\frac{1}{2^{2k+2}}\\ &\geq \frac{1}{4}+\frac{1}{2^{k+2}}+\frac{1}{2^{k+3}}-\frac{1}{2^{2k+2}}.\end{aligned}$$

If the remaining $\frac{1}{2^{k+3}} - \frac{1}{2^{2k+2}}$ yields a non-negative number, then we can simply subtract it from the right side without flipping the inequality.

Since k is a natural number, we have

$$k \geq 1.$$

Then, adding $k + 2$ to both sides gives

$$2k + 2 \geq k + 3.$$

Taking the reciprocal, we have

$$\frac{1}{k+3} \geq \frac{1}{2k+2} \quad \text{and} \quad \frac{1}{k+3} - \frac{1}{2k+2} \geq 0.$$

Subtracting $(\frac{1}{k+3} - \frac{1}{2k+2})$ from the RHS, we are left with

$$\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)...\left(1-\frac{1}{2^k}\right)\left(1-\frac{1}{2^{k+1}}\right) \geq \frac{1}{4}+\frac{1}{2^{k+2}}.$$

Hence $S(k + 1)$ is true. Thus, we have shown that $S(k)$ implies $S(k + 1)$.

Therefore, by the Principle of Mathematical Induction, $S(n)$ is true for $n \in \mathbb{N}$, establishing the inequality

$$\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)...\left(1-\frac{1}{2^n}\right) \geq \frac{1}{4}+\frac{1}{2^{n+1}}.$$

Supplemental Induction, Question 6:

For each natural number n, let $S(n)$ denote the statement that

$$1 + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{n}} \leq 2\sqrt{n} - 1.$$

(Basis Step) Then, $S(1)$ states $1 \leq 2\sqrt{1} - 1$. This simplifies to $1 \leq 1$, so $S(1)$ is true.

(Inductive Step) For an arbitrary natural number k, assume $S(k)$. That is,

$$1 + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{k}} \leq 2\sqrt{k} - 1.$$

Adding $\frac{1}{\sqrt{k+1}}$ to both sides, we have

$$1 + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq 2\sqrt{k} - 1 + \frac{1}{\sqrt{k+1}}. \tag{1}$$

This is equation 1. We will come back to it.

First, notice that since k is strictly positive, we have

$$\sqrt{k+1} \geq \sqrt{k}.$$

Adding $\sqrt{k+1}$ to both sides, we have

$$2\sqrt{k+1} \geq \sqrt{k+1} + \sqrt{k}.$$

Taking the reciprocal of both sides gives

$$\frac{1}{2\sqrt{k+1}} \leq \frac{1}{\sqrt{k+1} + \sqrt{k}}.$$

Multiplying both sides by $\frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k+1}-\sqrt{k}}$, we are left with

$$\frac{1}{2\sqrt{k+1}} \leq \sqrt{k+1} - \sqrt{k}.$$

Multiplying both sides by 2, we have

$$\frac{2}{2\sqrt{k+1}} \leq 2\sqrt{k+1} - 2\sqrt{k}. \tag{2}$$

Then, this is equation 2.

Now, consider the term $\frac{1}{\sqrt{k+1}}$. Multiplying it by $\frac{2}{2}$, we have

$$\frac{1}{\sqrt{k+1}} = \frac{2}{2\sqrt{k+1}}.$$

Substituting into $\frac{1}{\sqrt{k+1}}$ into equation 2, we have

$$\frac{1}{\sqrt{k+1}} \leq 2\sqrt{k+1} - 2\sqrt{k}.$$

Then, returning to equation 1, it follows that

$$\begin{aligned} 1 + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{k}} &\leq 2\sqrt{k} - 1 + 2\sqrt{k+1} - 2\sqrt{k} \\ &\leq 2\sqrt{k+1} - 1. \end{aligned}$$

Hence $S(k+1)$ is true. Thus, we have shown that $S(k)$ implies $S(k+1)$.

Therefore, by the Principle of Mathematical Induction, $S(n)$ is true for $n \in \mathbb{N}$, establishing the inequality

$$1 + \frac{1}{\sqrt{2}} + ... + \frac{1}{\sqrt{n}} \leq 2\sqrt{n} - 1.$$

Supplemental Induction, Question 8:

For each natural number n, let $S(n)$ denote the statement $3F_{n+1} + 2F_n = F_{n+4}$, where F_n denotes the Fibonacci Sequence $F_{n+2} = F_{n+1} + F_n$.

(Basis Step) Then, $S(1)$ states

$$3F_2 + 2F_1 = F_5.$$

Evaluating this, we have

$$3 \cdot 1 + 2 \cdot 1 = 5.$$

This simplifies to $5 = 5$, so $S(1)$ is true.

(Inductive Step) For an arbitrary natural number k, we assume $S(k)$. That is,

$$3F_{k+1} + 2F_k = F_{k+4}.$$

Adding F_{k+3} to both sides, we have

$$F_{k+3} + 3F_{k+1} + 2F_k = F_{k+3} + F_{k+4}.$$

Using the Fibonacci Recurrence by combining $F_{k+3} + F_{k+4}$ into F_{k+5} and expanding F_{k+3} into $F_{k+2} + F_{k+1}$, this reduces to

$$F_{k+2} + 4F_{k+1} + 2F_k = F_{k+5}.$$

We group $F_{k+1} + F_k$, giving us

$$F_{k+2} + 2F_{k+1} + 2(F_{k+1} + F_k) = F_{k+5}.$$

Then, combining $F_{k+1} + F_k$ into F_{k+2}, we have

$$3F_{k+2} + 2F_{k+1} = F_{k+5}.$$

Hence, $S(k+1)$ is true. Thus, we have shown that $S(k)$ implies $S(k+1)$.

Therefore, by the Principle of Mathematical Induction, $S(n)$ is true for $n \in N$, establishing the identity

$$3F_{n+1} + 2F_n = F_{n+4}.$$